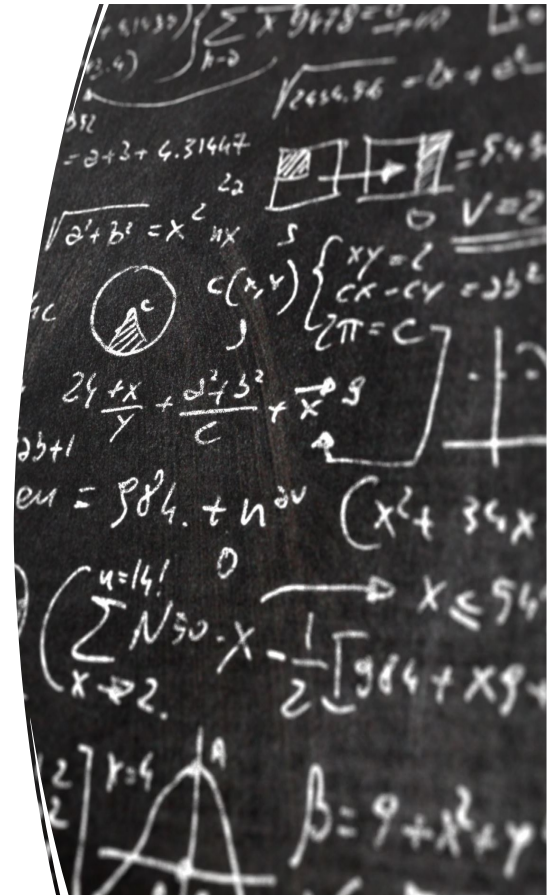


Statistical Modelling and Inference

Introduction

- Yu Zhang
- School of Mathematics Science
- Ocean University of China



About your Instructor

- 张侯 数学科学学院
- University of Minnesota- twin cities
- The University of Texas at Dallas
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Course Structure



满44减2 满88减5 满158减10

Lectures

- One lecture per week:

Week 1-4 8-16 Mon

8-10 Mon

- All lecture slides will be posted on BB.
- Theoretical Lecture
- Practical Lecture

Overview

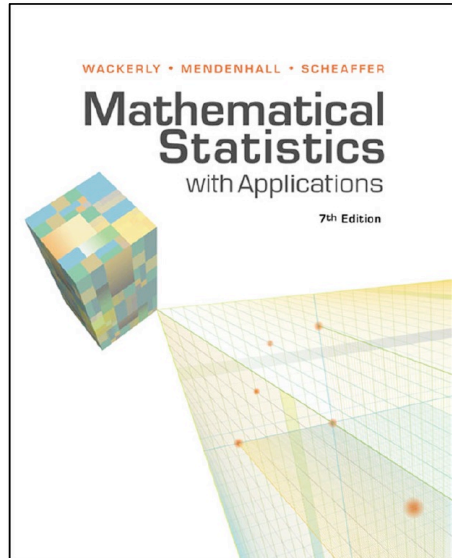
	Lectures	Test	Assessment
Week 1	Introduction R review		
Week 2	Bias, MSE, BLUE Practical 1 Data cleaning		
Week 3	Confidence Intervals, Hypothesis testing, Error and Power		Assignment 1 release
Week 4	P-values, Distributions, Sampling distribution Practical 2 Manipulating data		Assignment 1 due.
Week 5	Distributions of sample mean Distributions of sample variance, t and F distribution, One-sample t-test,		
Week 6	Inference for variance Two-sample t-test Practical 3 Graphics		Assignment 2 release
Week 7	SLR estimation SLR distributional results SLR hypothesis-tests		Assignment 2 due
Week 8	SLR prediction SLR residuals MLR setup		
Week 9	MLR estimation Practical 4 Rmarkdown Exam review		
Week 10	MLR properties of LSE MLR Inference MLR contrasts	Theory test	
Week 11	MLR hypothesis-tests Model selection_forwards Model selection_backwards Model selection_stepwise <u>Model selection_criteria</u>		Assignment 3 release

Week 12	Polynomial regression, Transformations <u>Practical 5 Linear models</u>		Assignment 3 due
Week 13	t-test MLR One way ANOVA MLR <u>Two way ANOVA MLR</u>		
Week 14	<u>ANCOVA</u> Maximum likelihood, Fisher information,		Assignment 4 release
Week 15	properties-of-score MLE confidence intervals Practical 6: ANCOVA:	Practical test	Assignment 4 due
Week 16	Final review		

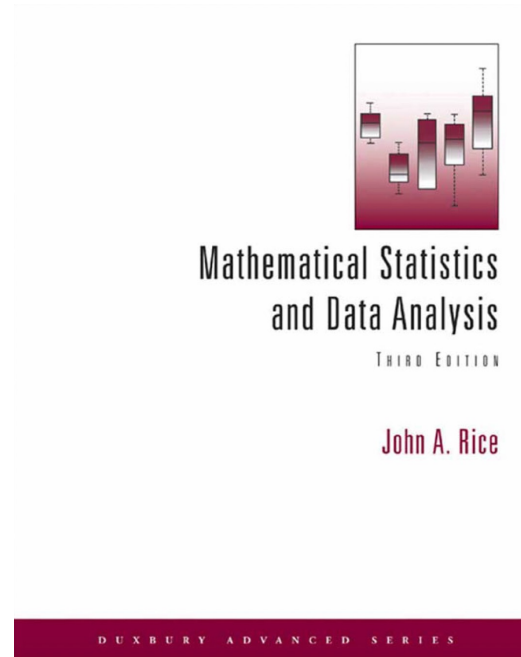
The course is structured into six modules:

1. Estimation Theory
2. Sampling Distributions
3. Linear Models I: Simple and Multiple Linear Regression
4. Linear Models II: Model selection, ANOVA, ANCOVA
5. Likelihood Theory
6. Practical Issues in Data Analysis

Reference text



Wackerley, DD, Mendenhall, W & Scheaffer, RL 2008, Mathematical statistics with applications, 7th edn, Duxbury Press



J. A. Rice: Mathematical Statistics and Data Analysis. Third edition (2007).

Grading
Policy

Assessment Task	Weighting
Attendance	10%
Assignments	20%
Theory Test	15%
Practical Test	15%
Final Exam	40%

Assignments

Assignment 1
due end of Week 4

Assignment 2
due end of Week 7

Assignment 3
due end of Week
12

Assignment 4
due end of Week
15

Release online at least 7 days before the due day.

Tests

Theoretical Test

Week 10

Practical Test

Week 15

OPEN BOOK, OPEN NOTES!

15% each

Final Exam

Date: to be
announced

40%

Distribution review

Notations

$$\begin{cases} H & f(H) = 1 \\ T & f(T) = 0 \end{cases}$$

- We use uppercase letters, X, Y, Z to denote random variables (RVs).
- We use lowercase letters x, y, z to represent real numbers.

X 正面朝上

$$X = \begin{cases} 0 \\ 1 \end{cases}$$

Discrete distributions

- The *probability mass function (PMF)* of a *discrete* random variable X is denoted by

$$p(x) = P(X = x)$$

- Some important examples are:
 - **Bernoulli Distribution**
 - **Binomial Distribution**
 - **Geometric Distribution**
 - **Poisson Distribution**

- ↗ parameter.
- **Bernoulli Distribution** $p(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases} \quad X \sim B(1, p)$

- Mean : p

- Variance: $p(1 - p)$

$$E(X) = \sum_{i=1}^{\infty} x_i p(x_i) = 0(1-p) + 1(p) = p$$

$$\text{Var}(X) = E(X - E(X))^2 = E(X^2) - E^2(X)$$

$$E(X^2) = \sum_{i=1}^{\infty} x_i^2 p(x_i)$$

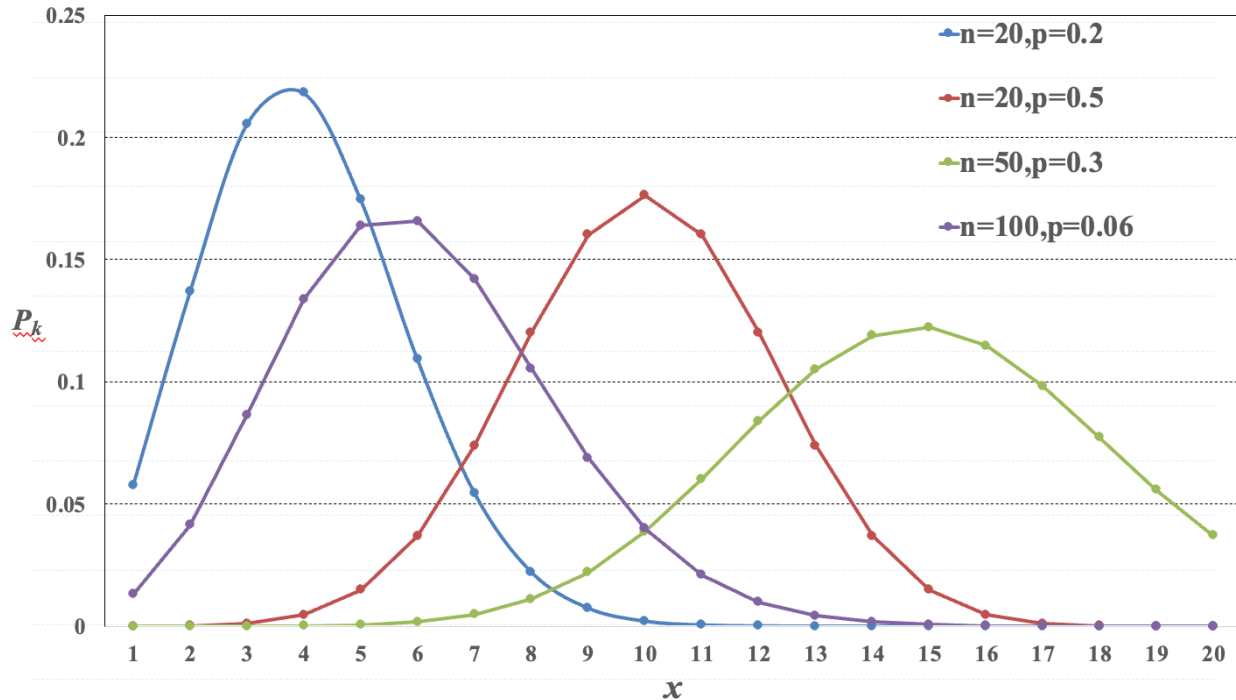
↗ $\frac{n!}{x!(n-x)!} = C_n^x$

Binomial Distribution $p(x) = \binom{n}{x} p^x (1-p)^{n-x}$, for $x = 0, 1, \dots, n$

Mean : np

$$X \sim B(n, p)$$

Variance: $np(1-p)$



- **Geometric Distribution** $p(x) = p (1 - p)^{x-1}$, for $x = 1, 2, \dots$
- The probability distribution of the number X of [Bernoulli trials](#) needed to get one success
- Mean : $\frac{1}{p}$
- Variance: $\frac{1-p}{p^2}$

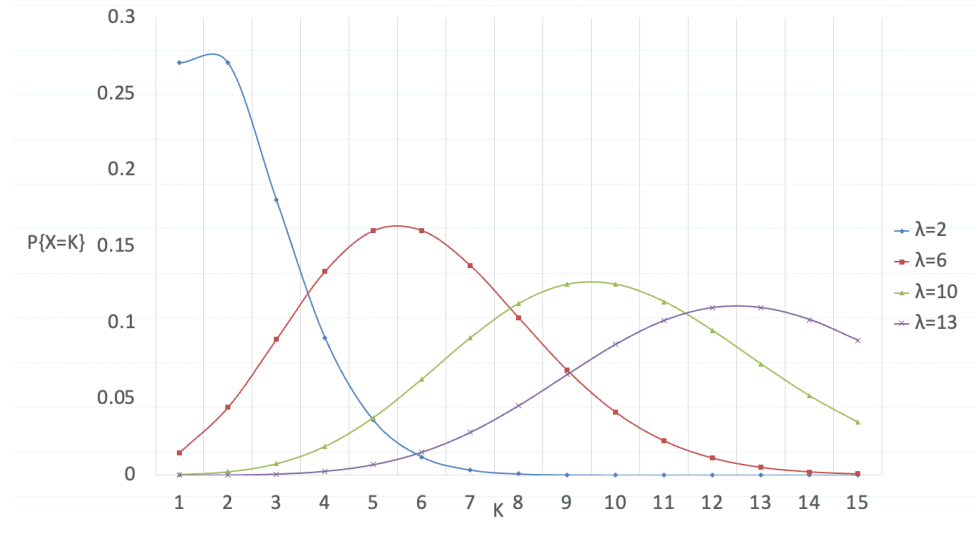
Poisson Distribution $p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$

to predict the probability of certain events happening when you know how often the event has occurred

$$X \sim \text{Po}(\lambda)$$
$$\pi(\lambda)$$

Mean: λ

Variance: λ



$$\uparrow \downarrow$$
$$n \cdot p = \lambda$$

$$\lim_{n \rightarrow \infty} \binom{n}{k} p^k (1-p)^{n-k} = \frac{\lambda^k e^{-\lambda}}{k!}$$

Cumulative distribution function

$$F(x) = P\{X \leq x\}$$

PMF

$$p(x) = P(X = x)$$



CDF

$$F(x) = \sum_{x_i \leq x} p(x_i)$$

$$(1) \quad x_1 \leq x_2 \quad F(x_1) \leq F(x_2)$$

$$(2) \quad 0 \leq F(x) \leq 1$$

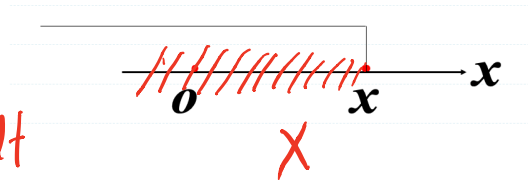
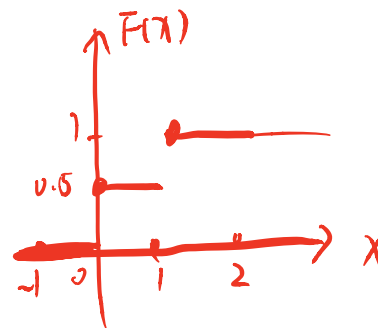
$$F(-\infty) = 0 \quad F(+\infty) = 1$$

(3) 右连续

pdf
 $f(x)$



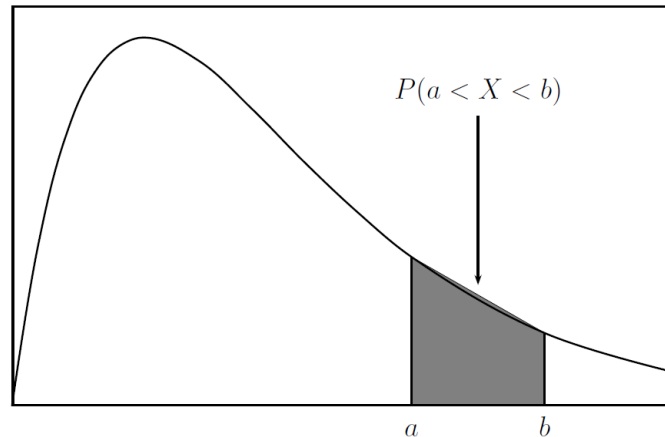
$$F(x) = \int_{-\infty}^x f(t) dt$$



Continuous distributions

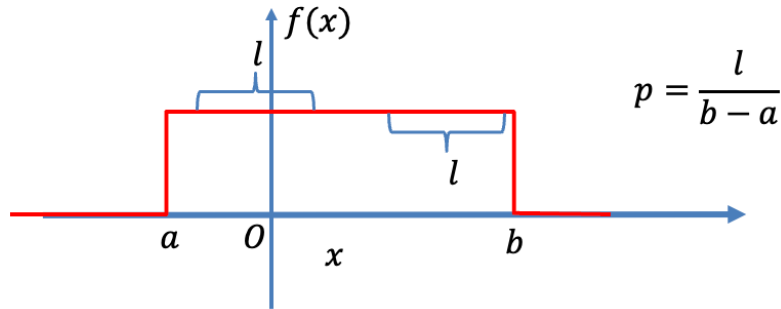
- The *probability density function (PDF)* of a *continuous* random variable X is denoted by $f(x)$. It uses area to represent probability.

$$P(a \leq x \leq b) = \int_a^b f(x) dx$$



Continuous distributions $X \sim U(a, b)$

- **Uniform Distribution** $f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise.} \end{cases}$
- Mean: $\frac{a+b}{2}$
- Variance: $\frac{(b-a)^2}{12}$



Exponential Distribution $f(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta}, & x > 0, \\ 0, & x \leq 0. \end{cases}$ $X \sim E(\theta)$

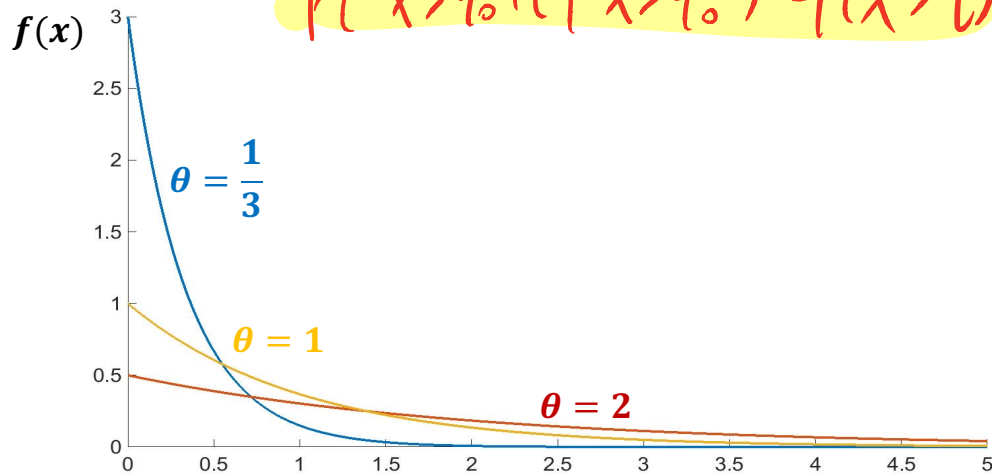
Mean: θ $\frac{1}{\lambda}$

Variance: θ^2 $\frac{1}{\lambda^2}$

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$X \sim E(\lambda)$

$$P(X > t_0 + t | X > t_0) = P(X > t)$$

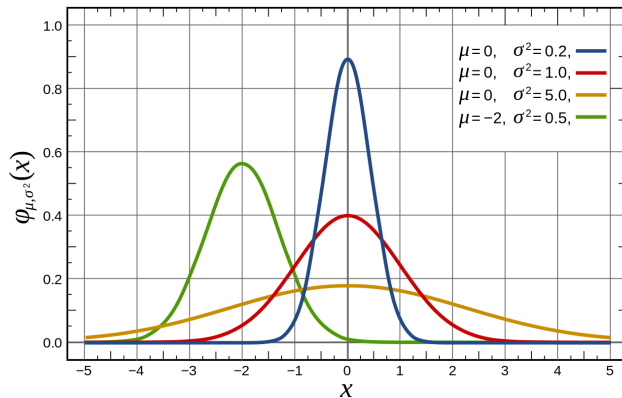


- Normal Distribution $N(\mu, \sigma^2)$

$$X \sim N(\mu, \sigma^2)$$

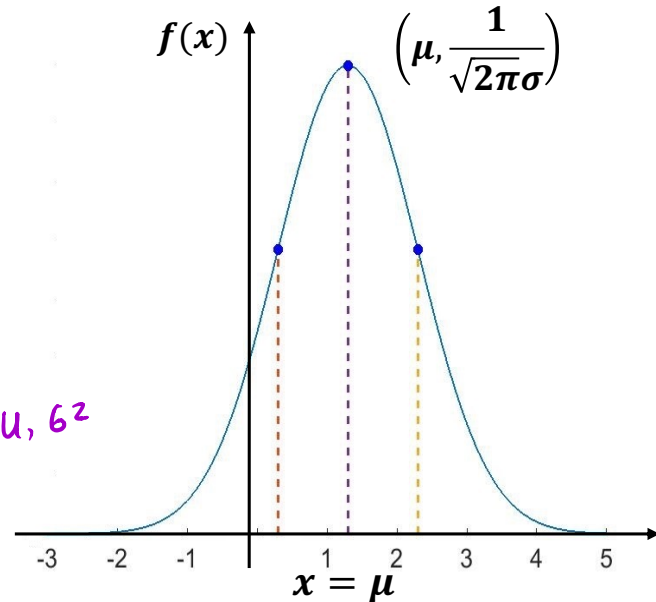
$$\begin{matrix} \downarrow & \downarrow \\ E(X) & \text{Var}(X) \end{matrix}$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



$$f(x) \quad \left(\mu, \frac{1}{\sqrt{2\pi}\sigma} \right)$$

μ, σ^2



$$P(a < X < b) = \int_a^b f(x) dx$$

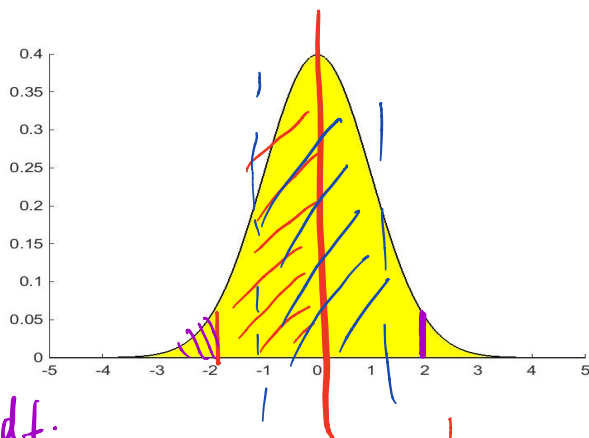
$$= P\left(\frac{a-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

- Standard Normal Distribution $N(0, 1)$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$\psi(z) \quad \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty < z < \infty,$$

$$\Phi(z) \quad \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt, \quad -\infty < z < \infty.$$

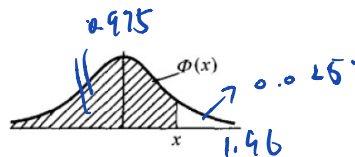


cdf.

$$\Phi(0) = P(X \leq 0) = \frac{1}{2}$$

$$\Phi(-x) = 1 - \Phi(x)$$

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$



x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936